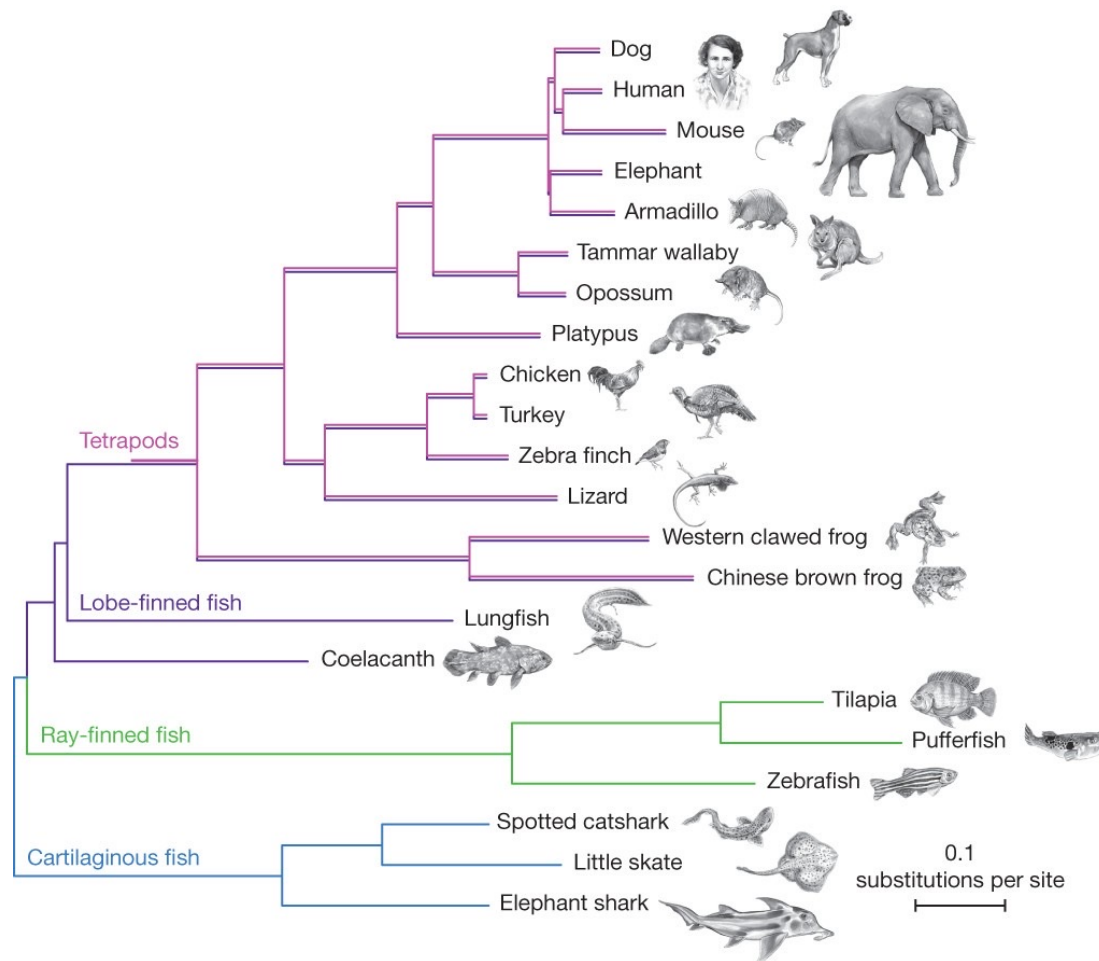


Phylogenetics



ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC

Locus 1

TTGCAAAGGGGGCGCGGGCC
TTGCAAAGGGGGCGCGGGCC
TTGCAAAGGGGGCGCGGGCC
TTGCAAAGGGGGCGCGGGCC
TTGCAAAGGGGGCGCGGGCC

Locus 2

...

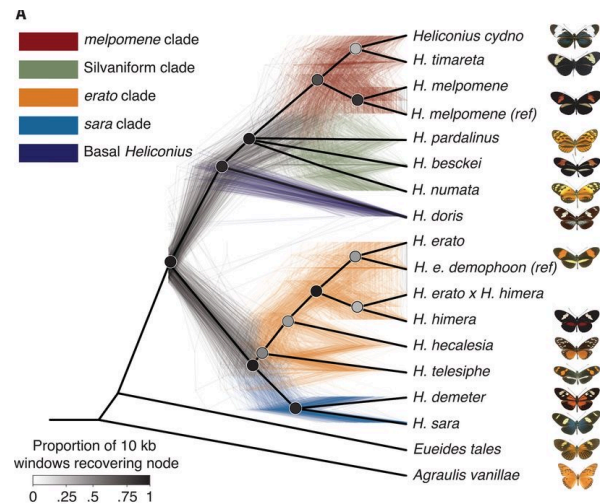
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA

Locus *n*



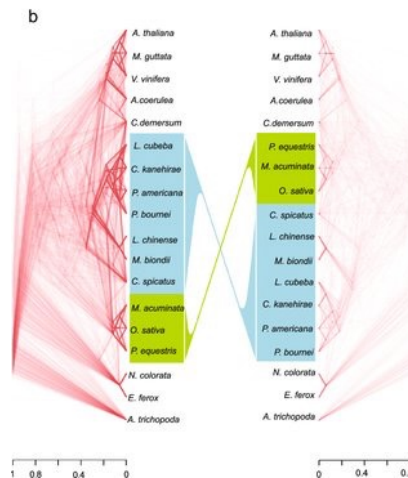
Genomic era

Heliconius



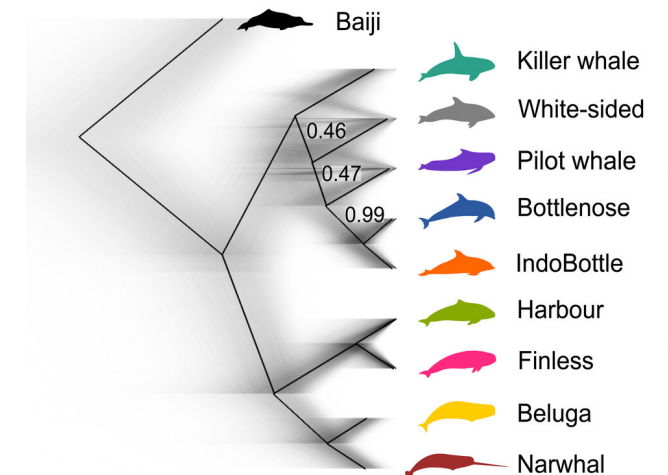
Edelman et al. 2019 Science

Mesangiosperm



Guo et al. 2021 Nature Comm.

Delphinoidea



Wesbury et al. 2023 Mol. Ecol.

Technical causes: not related to the evolutionary history, but to technical problems that leads to **wrong estimation of trees**

- Selection of genes
- Sequencing errors
- Taxon ID
- Different human errors

Gene processes: related to non-independent evolutionary history

- Ortholog/paralog confusion
- Evolutionary rate heterogeneity among taxa or sites
- Intralocus recombination
- Nucleotide composition bias

Species level process: real discordance related to different evolutionary histories

- Incomplete lineage sorting
 - Horizontal gene transfer / introgression
- ← Multispecies coalescent

Phylogenetic discordance: incomplete lineage sorting and coalescence

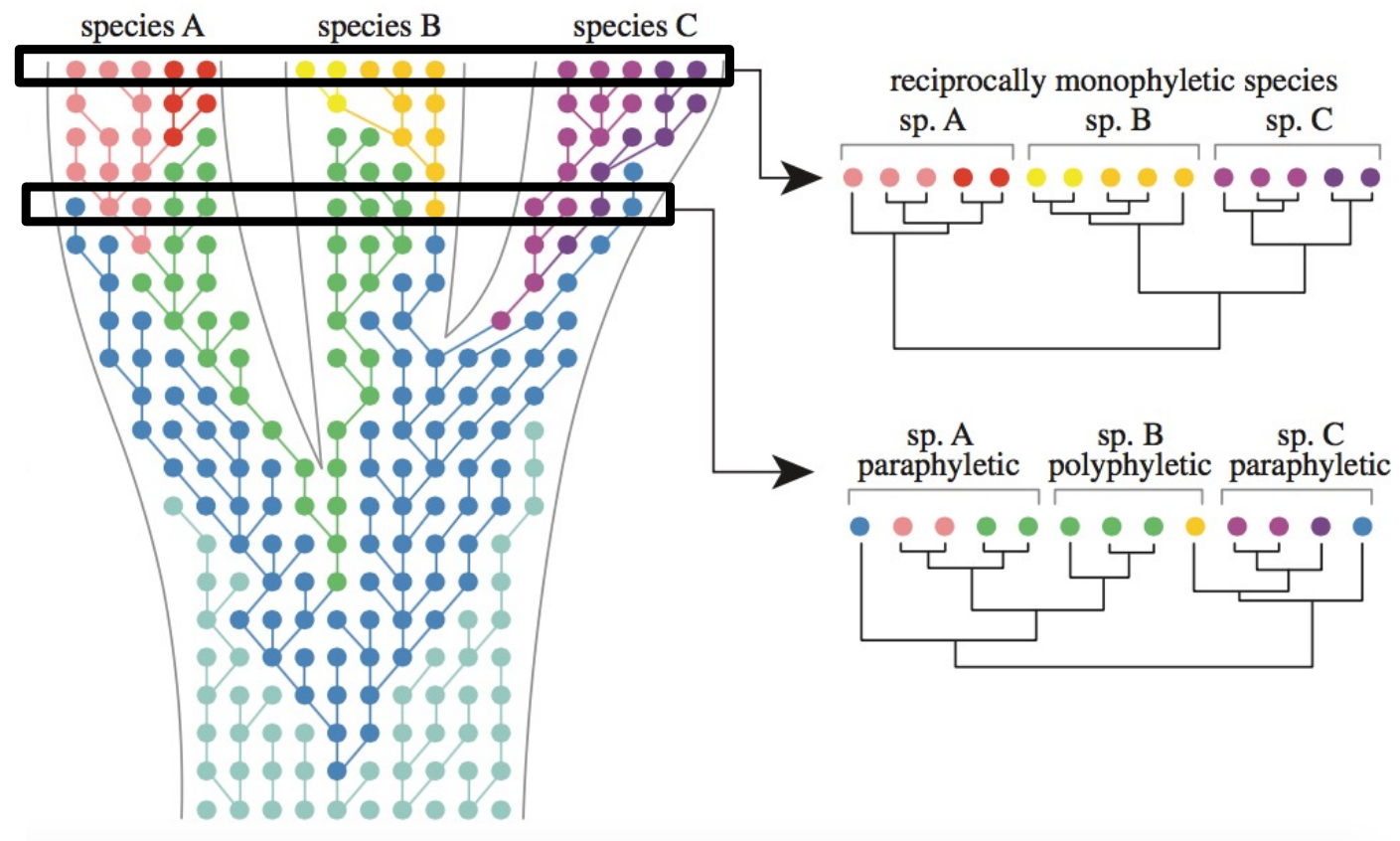
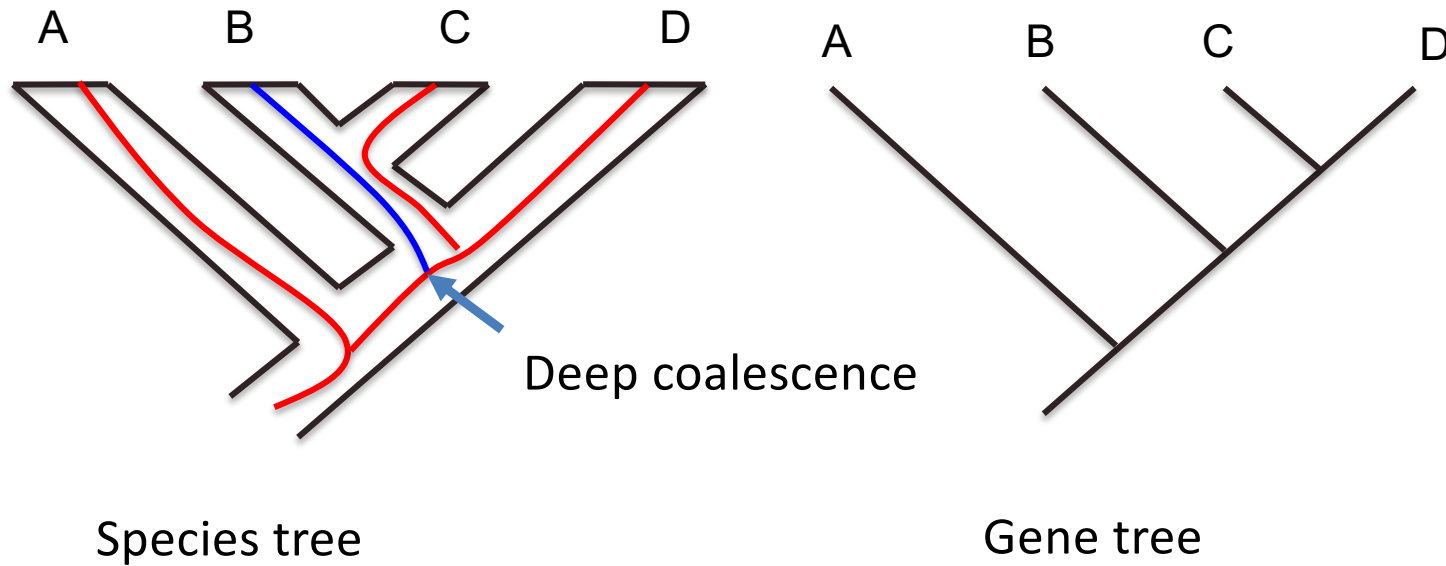
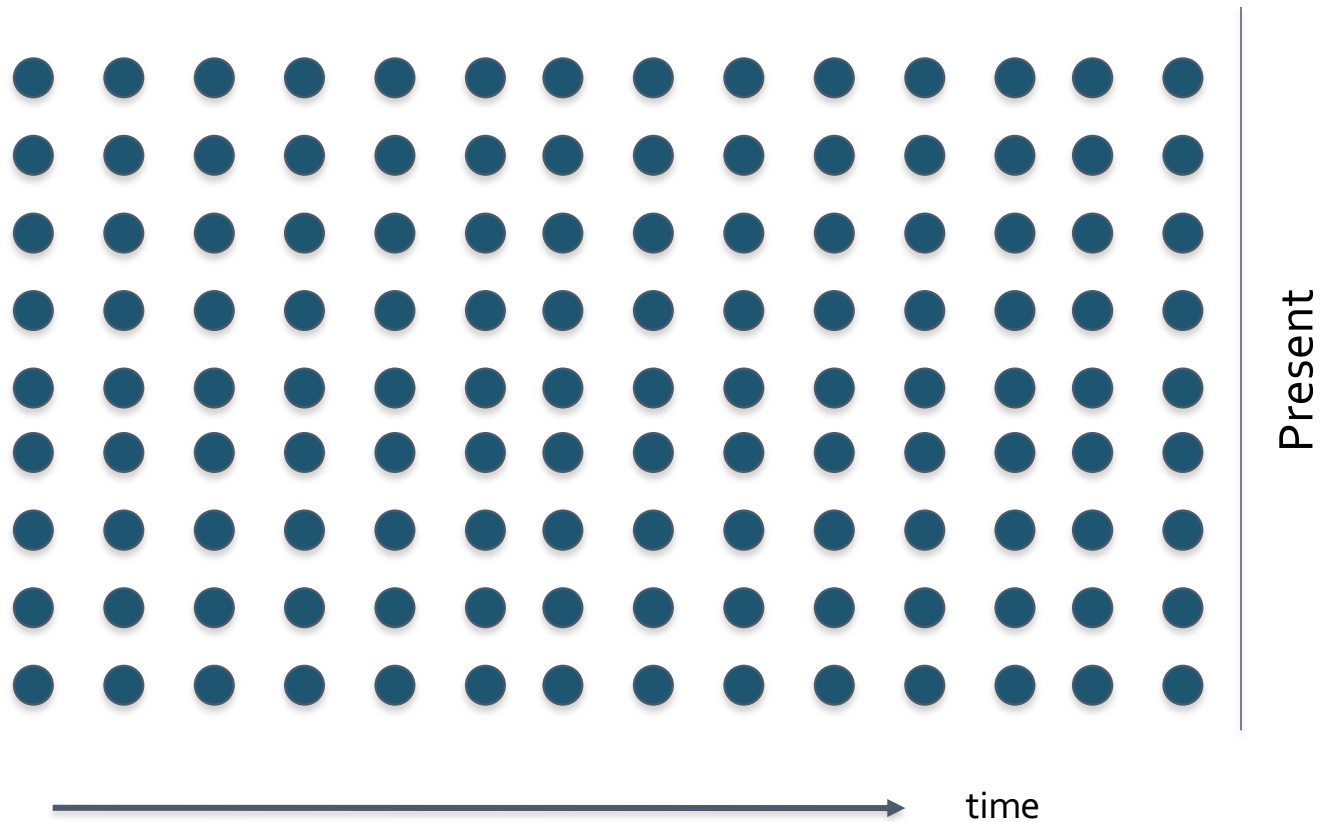


Fig. de Leliaert et al. 2014

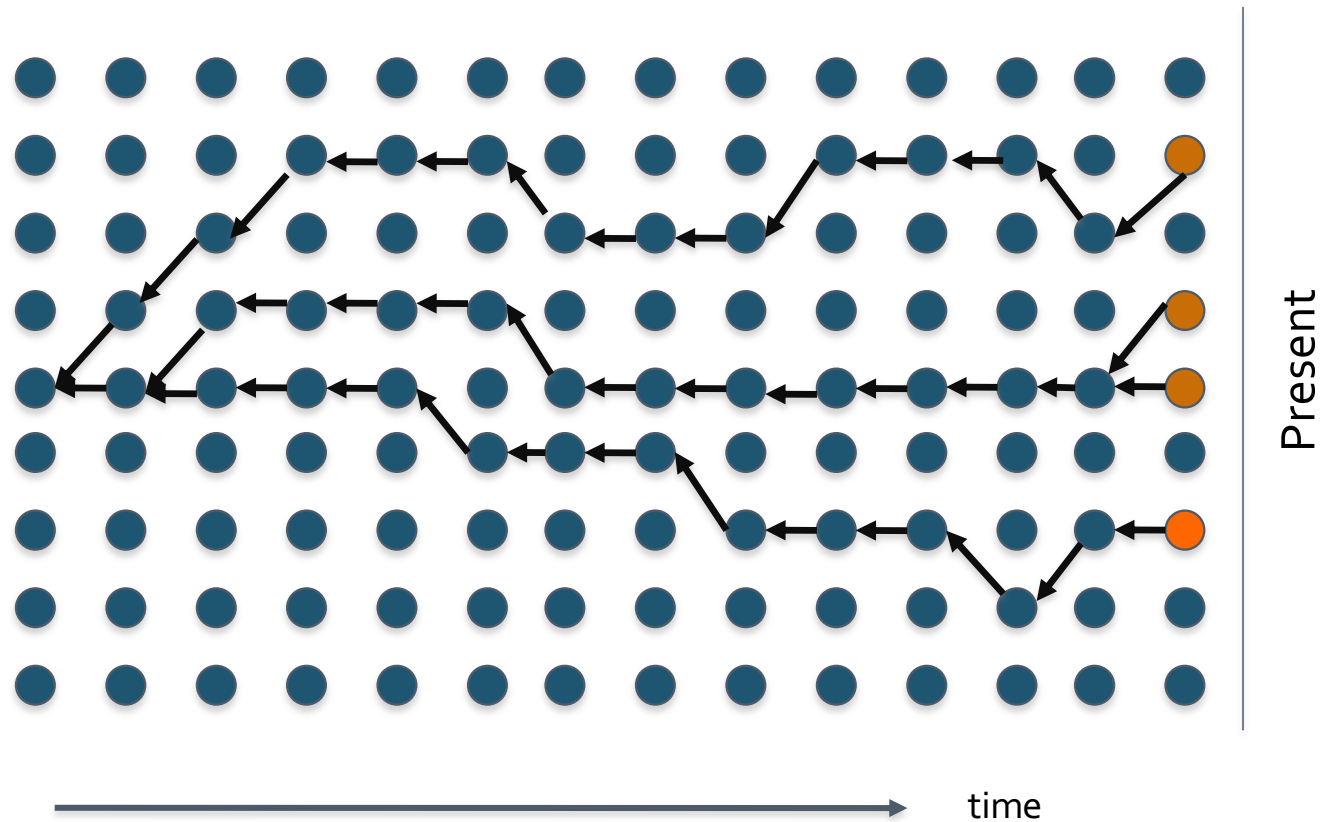
Incomplete Lineage Sorting (ILS)



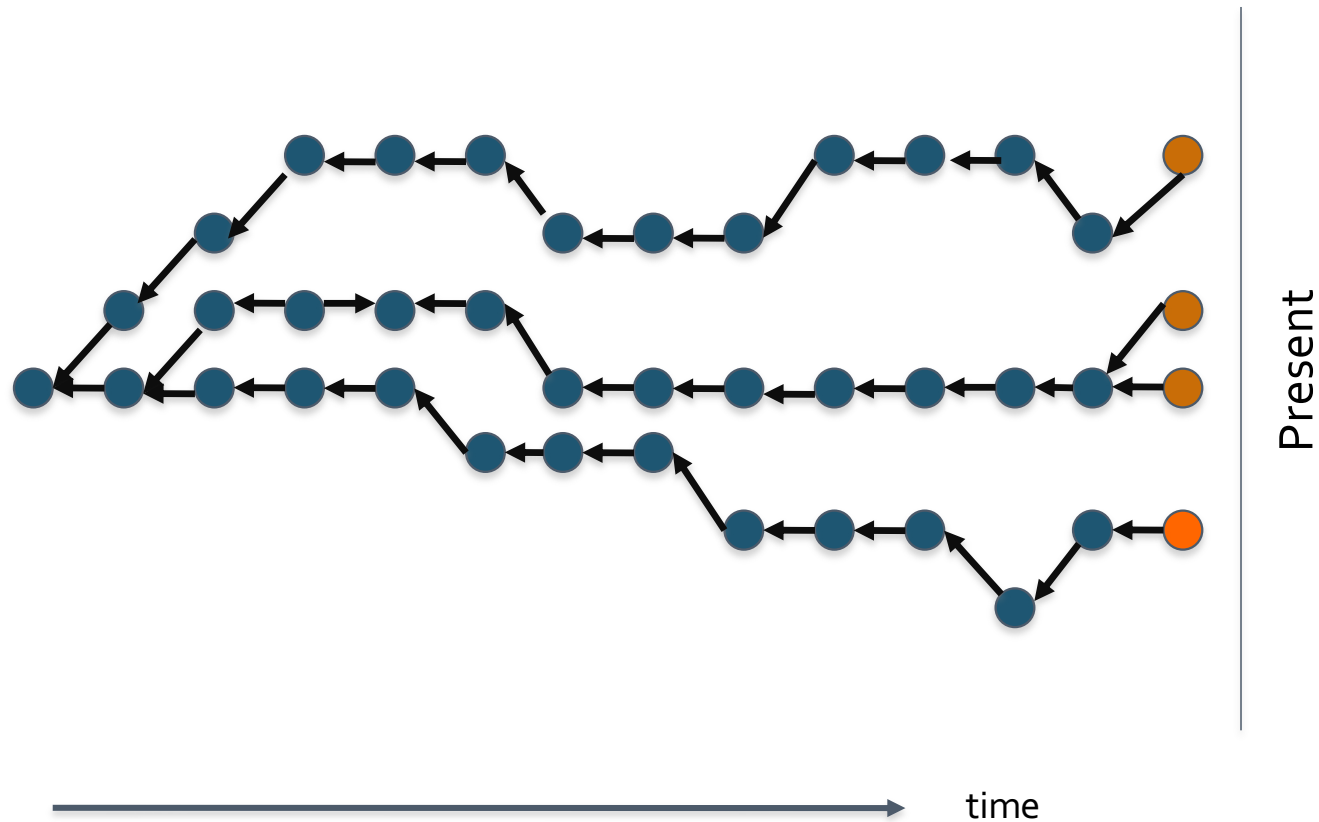
Population alleles



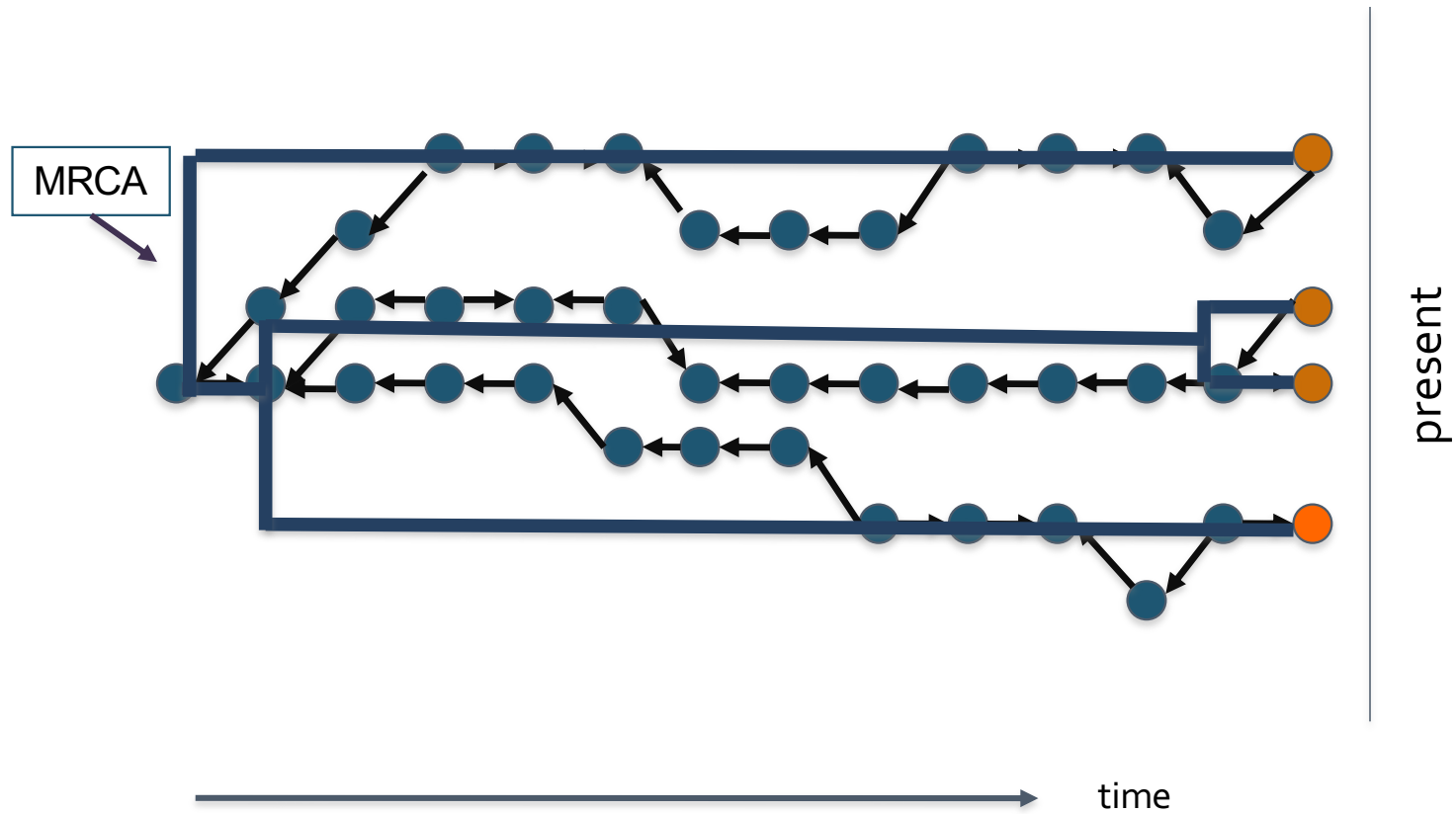
Population alleles



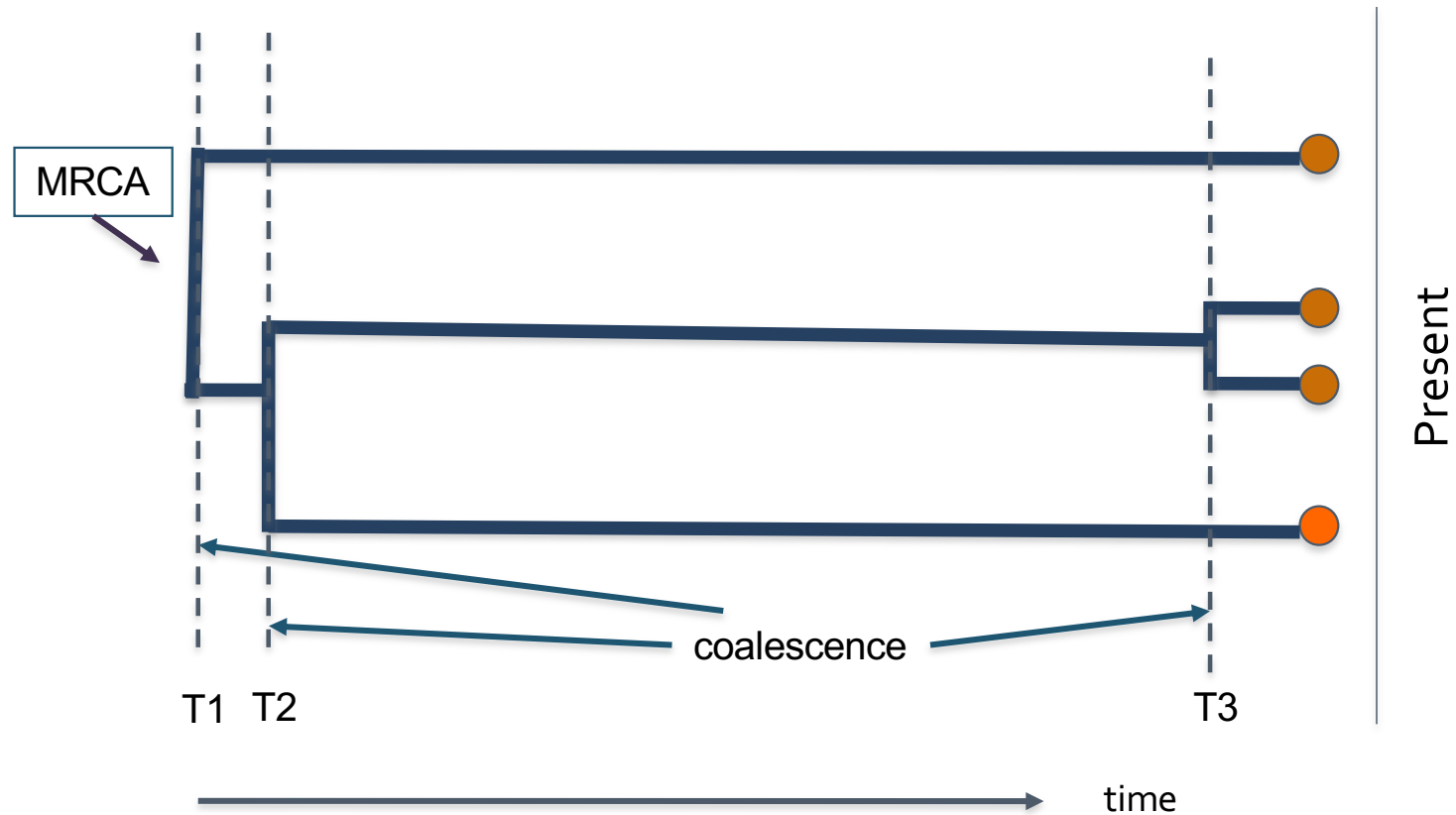
Population alleles



Coalescence

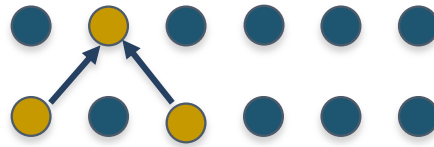


Coalescence



Coalescence

Diploid Eukaryotes



N_e = effective population size

$$P(\text{coal}, n=2) = \frac{1}{2N}$$

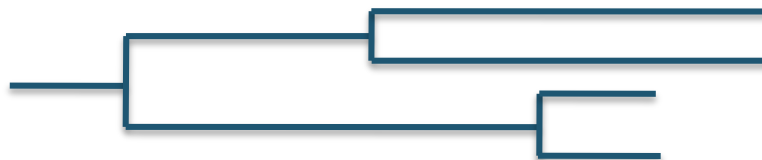
$$E(\text{coal}) = 2N_e$$

Coalescence

$$E(\text{coal}) = 2N$$

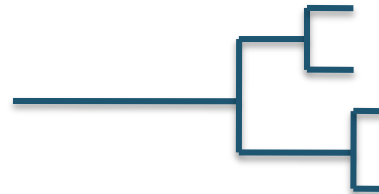


✓ Larger population size $\rightarrow$ longer coalescence time



Longer branches in gene trees

✓ Smaller population size $\rightarrow$ shorter coalescence time



Shorter branches in gene trees

Phylogenetic tree of Bivalvia showing age in Ma (x-axis, 800 to 0) and data used to place families (y-axis). The tree is rooted at 800 Ma and shows the divergence of various bivalve families. The data used to place families is indicated by colored bars on the right side of the tree, corresponding to the data types used in the analysis.

Legend for data used to place family:

- [T] Transcriptome
- [S] Sanger
- [L] Localized molecular
- [M] morphology

Legend for branch length changes:

- Blue bar: 270 Myr added to branch length in bifurcating model
- Grey bar: no change in branch length between models
- Red bar: 270 Myr shorter branch length in bifurcating model

Phylogenetic groups (clades) shown on the right:

- Protobranchia
- Pteriomorpha
- Archiheterodonta
- Anomalodesmata
- Imparidentia
- Palaeoheterodonta

Phylogenetic tree structure (families listed from top to bottom):

- Solemyidae
- Nuculidae
- Maclureidae
- Nuculanidae
- Paralelodontidae
- Noetidae
- Arctidae
- Cucullidae
- Cyclopteridae
- Phiolariidae
- Limosidae
- Limidae
- Enlidae
- Macuridae
- Picatulidae
- Omyidae
- Anomyidae
- Spondylidae
- Cycloclamyidae
- Propeamussidae
- Pectinidae
- Mytilidae
- Pinnidae
- Onchidae
- Chlamidae
- Malacidae
- Pendae
- Pulvinidae
- Isosommatidae
- Vulsellidae
- Margaritidae
- Trigonidae
- Malpandidae
- Unionidae
- Hyndae
- Mycetopodidae
- Indidae
- Ephraidae
- Carditidae
- Astartidae
- Crassatellidae
- Phacelomyidae
- Parimylidae
- Lyonsellidae
- Eucardiidae
- Spheniopsidae
- Cuspididae
- Latunellidae
- Pencilidae
- Caagellidae
- Pandidae
- Lyonsidae
- Poromyidae
- Cedpharidae
- Myochamidae
- Pionomidae
- Thracidae
- Thysanidae
- Lucinidae
- Gastrochlamidae
- Solecurtidae
- Psammobidae
- Donacidae
- Semellidae
- Cardidae
- Ellidae
- Solenidae
- Pholidae
- Basterotidae
- Lesaeidae
- Galeommatidae
- Sphaeridae
- Arctidae
- Phacelidae
- Corbulidae
- Dreissenidae
- Myidae
- Astartidae
- Macuridae
- Mesodermatidae
- Cymidae
- Gammaridae
- Unguidae
- Cyrenidae
- Cyrenoidae
- Glaucornidae
- Kellidae
- Vesicomyidae
- Arctidae
- Neoleptodontidae
- Veneridae
- Chamidae
- Hemidonacidae
- Trapezidae
- Glossidae

Concatenation

ATCAAAATTTGGCGGC	TTGCAAAGGGGCGCGGGCC	CCTGCTCCCGGTCGTA
ATCAAAATTTGGCGGC	TTGCAAAGGGGCGCGGGCC	CCTGCTCCCGGTCGTA
ATCAAAATTTGGCGGC	TTGCAAAGGGGCGCGGGCC	CCTGCTCCCGGTCGTA
ATCAAAATTTGGCGGC	TTGCAAAGGGGCGCGGGCC	CCTGCTCCCGGTCGTA
ATCAAAATTTGGCGGC	TTGCAAAGGGGCGCGGGCC	CCTGCTCCCGGTCGTA

Locus 1

Locus 2

Locus 3

- It forces the analysis to treat all genes as if they were a single “**mega gene**” that must represent the same phylogenetic tree.
- More informative loci (with more mutations) carry greater weight in the reconstruction.

Sequences

Locus 1

```
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
```

Locus 2

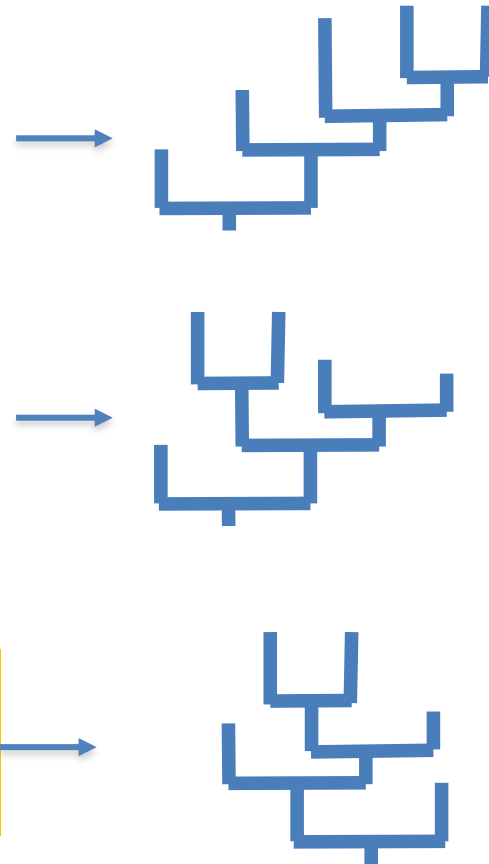
```
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
```

⋮

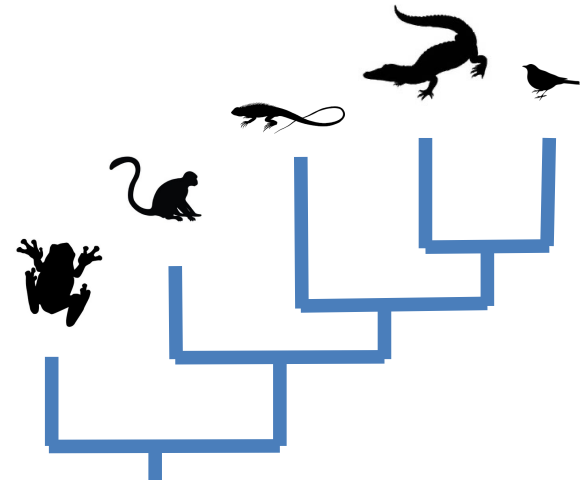
Locus n

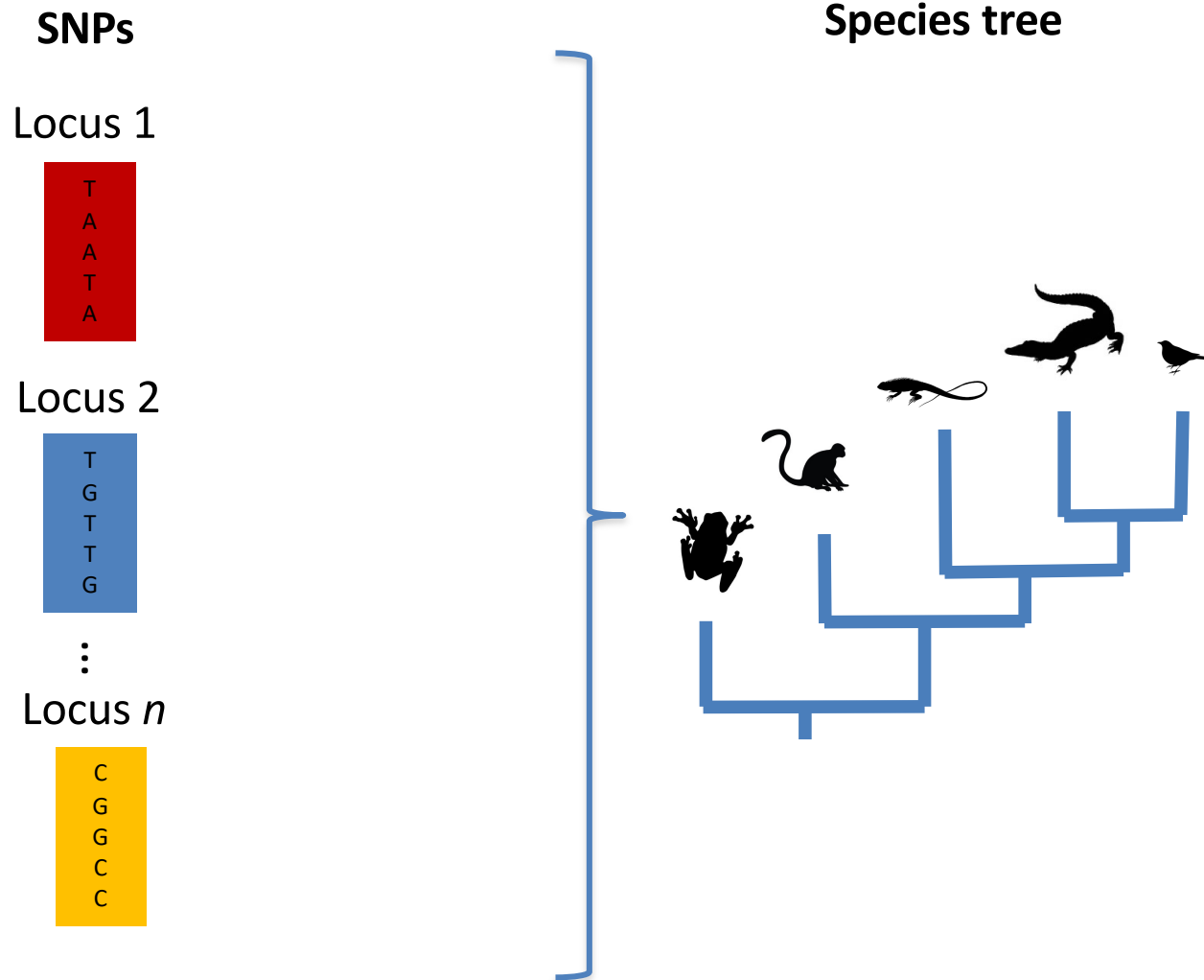
```
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
```

Independent gene trees



Species trees





Assumptions

Gene trees

- No technical errors in the sequences
- Genes are homologous
- No intralocus recombination

Species tree

- All gene trees meet the assumptions listed above
- Each gene tree / SNP is independent (no linkage)
- No natural selection acting on gene trees / SNPs
- Discordance among gene trees is caused only by incomplete lineage sorting

Programas para estimar árboles de especies

Program	Type of data	loci	Species / individuals	time consumption
starBEAST3	long sequences	hundreds	few	slow
SNAPPER	SNPs	hundreds	few	slow
MP-EST	gene trees	thousands	many	medium
ASTRAL	gene trees	thousands	many	fast
SVDquartets	SNPs	thousands	many	fast

Sequences

Locus 1

```
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
```

Locus 2

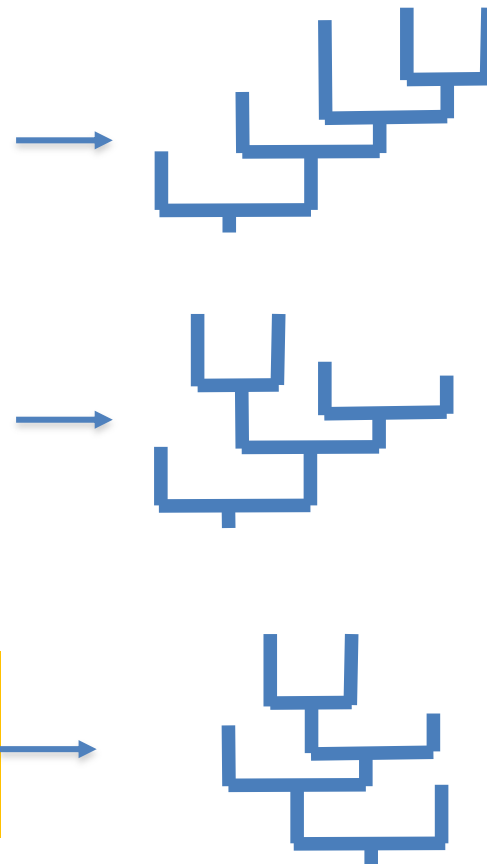
```
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
```

⋮

Locus n

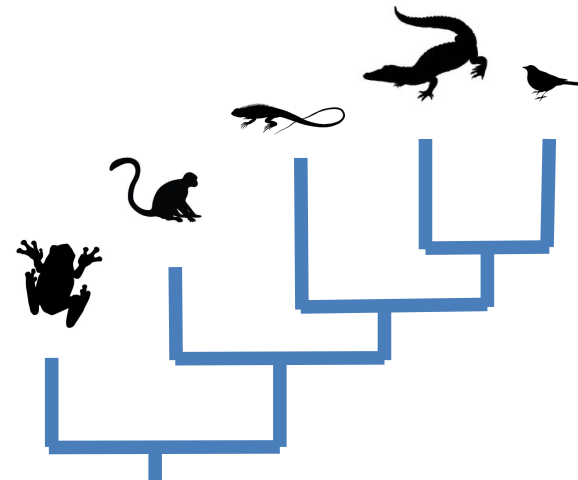
```
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
```

Independent gene trees

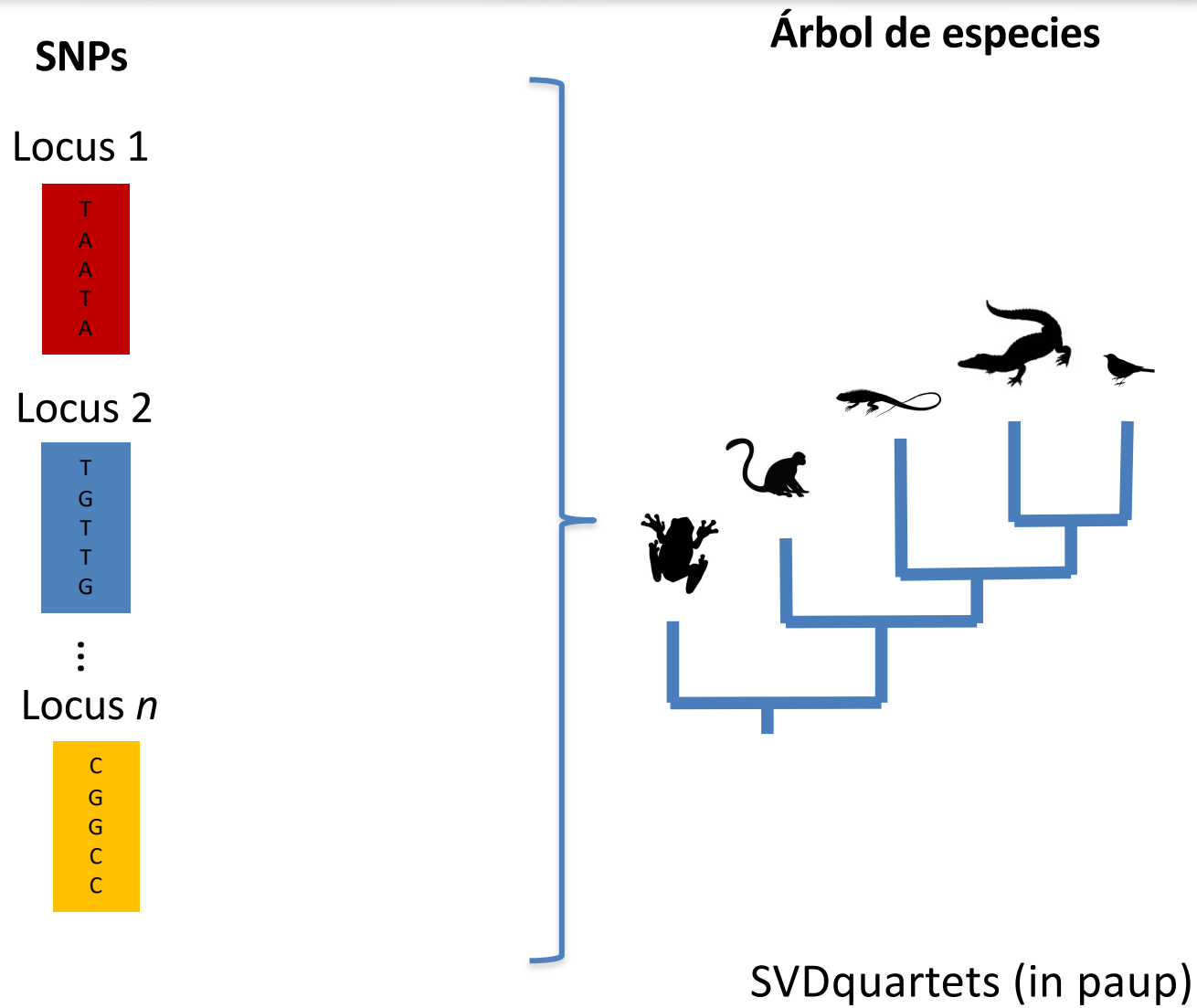


raxml

Species tree



ASTRAL



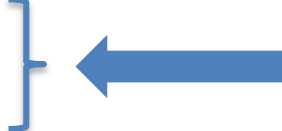
Technical causes: not related to the evolutionary history, but to technical problems that leads to **wrong estimation of trees**

- Selection of genes
- Sequencing errors
- Taxon ID
- Different human errors

Gene processes: related to non-independent evolutionary history

- Ortholog/paralog confusion
- Evolutionary rate heterogeneity among taxa or sites
- Intralocus recombination
- Nucleotide composition bias

Species level process: real discordance related to different evolutionary histories

- Incomplete lineage sorting
 - Horizontal gene transfer / introgression
- 
- Multispecies network coalescent

Influence of hybridization in species tree methods

Other

to gene tree / species

Syst. Biol. 63(1):17–30, 2014
© The Author(s) 2013. Published by Oxford University Press, on behalf of the Society of Systematic Biologists. All rights reserved.
For Permissions, please email: journals.permissions@oup.com
DOI:10.1093/sysbio/syt049
Advance Access publication August 13, 2013

The Influence of Gene Flow on Species Tree Estimation: A Simulation Study

ADAM D. LEACHÉ^{1,*}, REBECCA B. HARRIS¹, BRUCE RANNALA^{2,3}, AND ZIHENG YANG^{3,4}

¹Department of Biology and Burke Museum of Natural History and Culture, University of Washington, Seattle, WA 98195 USA;

²Genome Center and Department of Evolution & Ecology, University of California, Davis, CA 95616, USA;

³Center for Computational Genomics, Beijing Institute of Genomics, Chinese Academy of Sciences, Beijing 100101, China; and

Syst. Biol. 65(5):843–851, 2016
© The Author(s) 2016. Published by Oxford University Press, on behalf of the Society of Systematic Biologists. All rights reserved.
For Permissions, please email: journals.permissions@oup.com
DOI:10.1093/sysbio/syw030
Advance Access publication May 5, 2016

Inconsistency of Species Tree Methods under Gene Flow

CLAUDIA SOLÍS-LEMUS^{1,*}, MENGYAO YANG¹, AND CÉCILE ANÉ^{1,2}

¹Department of Statistics, University of Wisconsin, Madison, WI, 53706, USA; ²Department of Botany, University of Wisconsin, Madison, WI, 53706, USA

*Correspondence to be sent to: Claudia Solís-Lemus, Department of Statistics, University of Wisconsin-Madison, 1220 Medical Sciences Center, 1300 University Avenue, Madison, WI, 53706, USA; E-mail: claudia@stat.wisc.edu.

Syst. Biol. 0(0):1–17, 2018
© The Author(s) 2018. Published by Oxford University Press, on behalf of the Society of Systematic Biologists. All rights reserved.
For permissions, please email: journals.permissions@oup.com
DOI:10.1093/sysbio/syy020

The Effect of Gene Flow on Coalescent-based Species-Tree Inference

COLBY LONG^{1,*} AND LAURA KUBATKO²

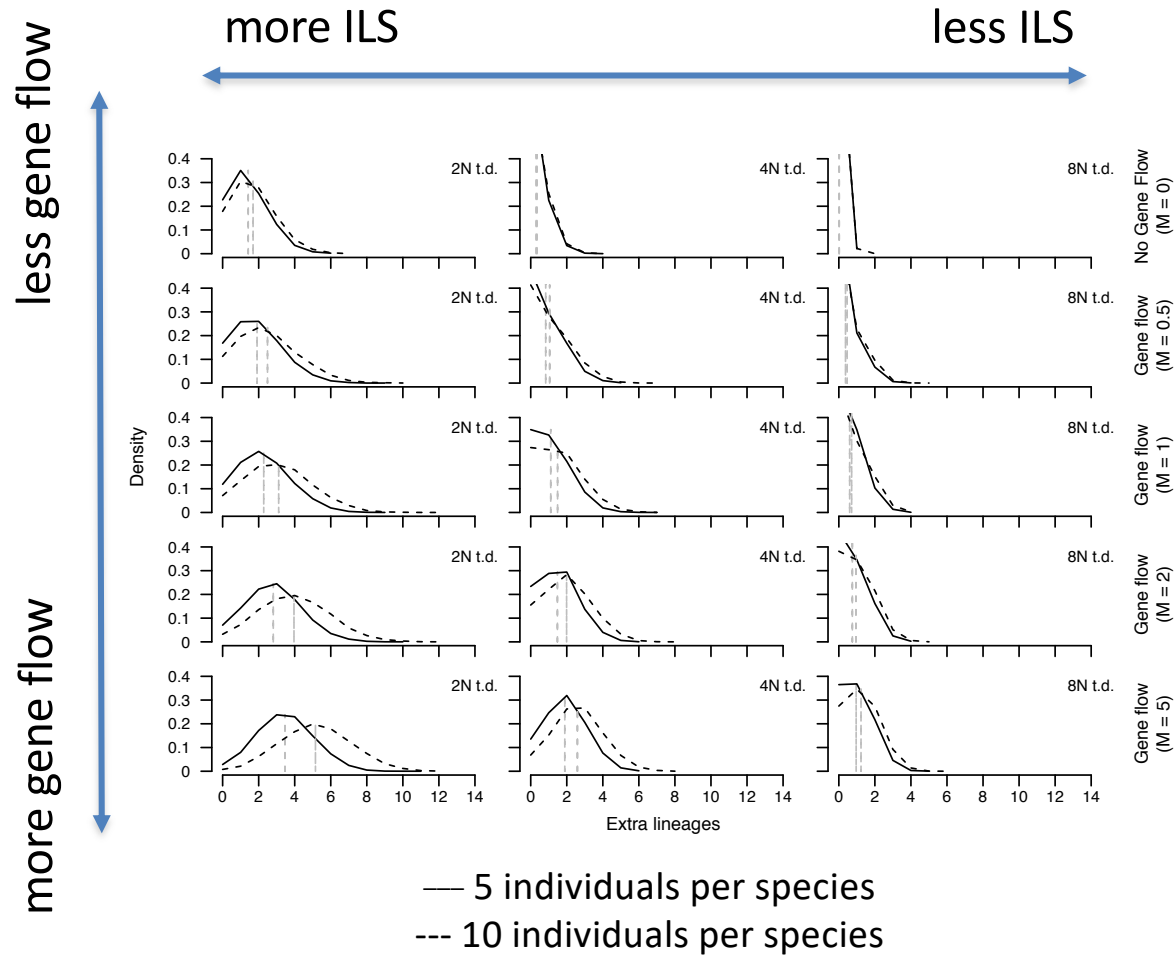
¹Mathematical Biosciences Institute, The Ohio State University, 1735 Neil Ave., Columbus, OH, USA and ²Department of Statistics, Department of Evolution, Ecology, and Organismal Biology, The Ohio State University, 1958 Neil Ave., Columbus, OH, USA

*Correspondence to be sent to: Mathematical Biosciences Institute, The Ohio State University, 1735 Neil Ave., Columbus, OH, USA; E-mail: long.1579@mbi.osu.edu.

Received 19 September 2017; reviews returned 5 March 2018; accepted 6 March 2018
Associate Editor: Cecile Ane

t. Biol.

Influence of hybridization in species tree methods



Assumptions

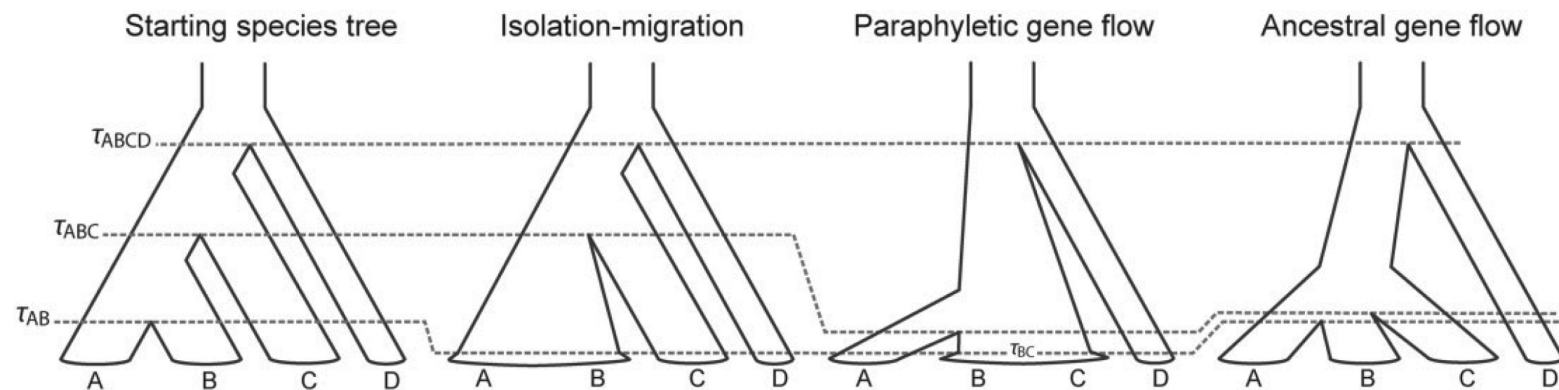
Gene trees

- No technical errors in the sequences
- Genes are homologous
- No intralocus recombination

Species tree (Multispecies Coalescent)

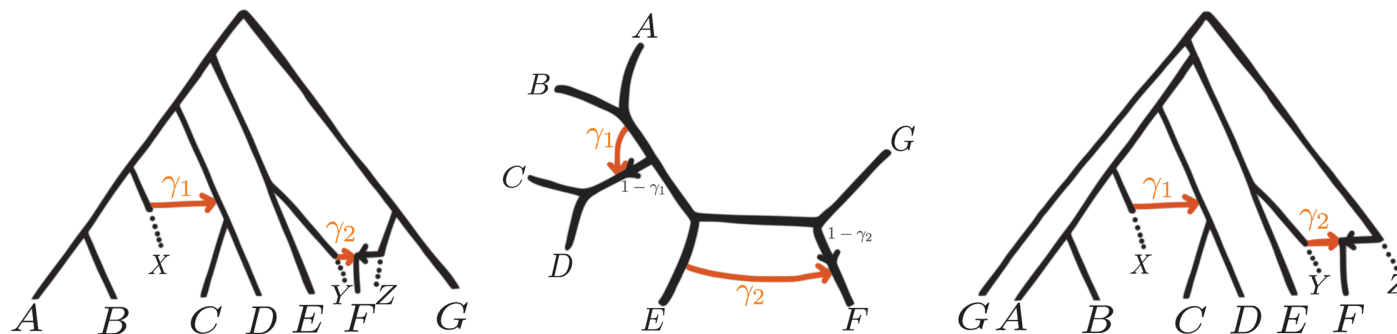
- All gene trees meet the assumptions listed above
- Each gene tree / SNP is independent (no linkage)
- No natural selection acting on gene trees / SNPs
- **Discordance among gene trees is caused only by incomplete lineage sorting**

Influence of hybridization in species tree methods



Leaché et al. 2014; Syst. Biol.

The Multispecies Network Coalescent (MSNC) accommodates ILS and gene flow



Phylogenetic networks are phylogenetic trees with added hybrid edges

Solis-Lemus et al. (2016); Plos Genetics
Blair and Ané (2020); Syst. Biol.

Assumptions

Gene trees

- No technical errors in the sequences
- Genes are homologous
- No intralocus recombination

Phylogenetic networks

- All gene trees meet the assumptions listed above
- Each gene tree / SNP is independent (no linkage)
- No natural selection acting on gene trees / SNPs
- ~~Discordance among gene trees is caused only by incomplete lineage sorting~~



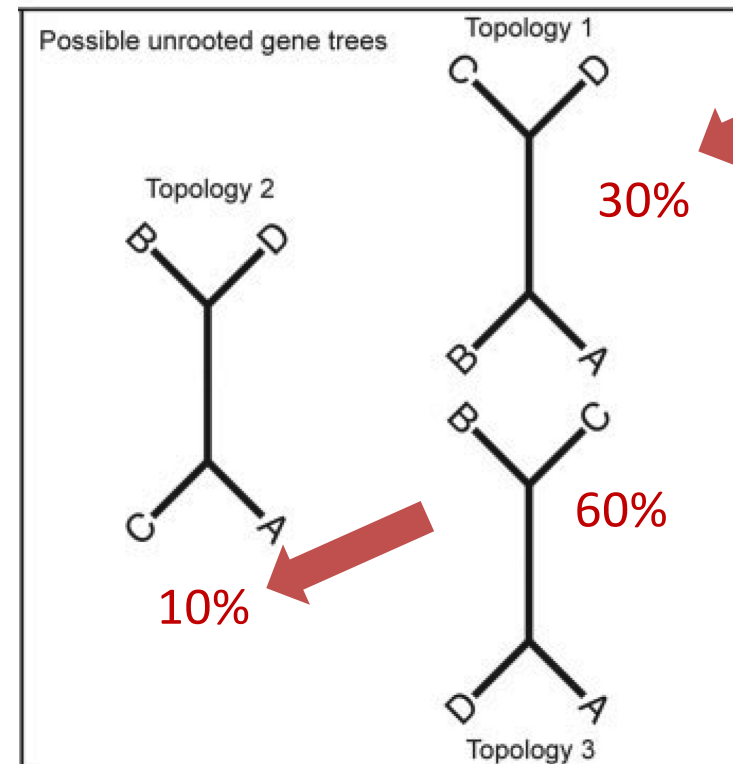
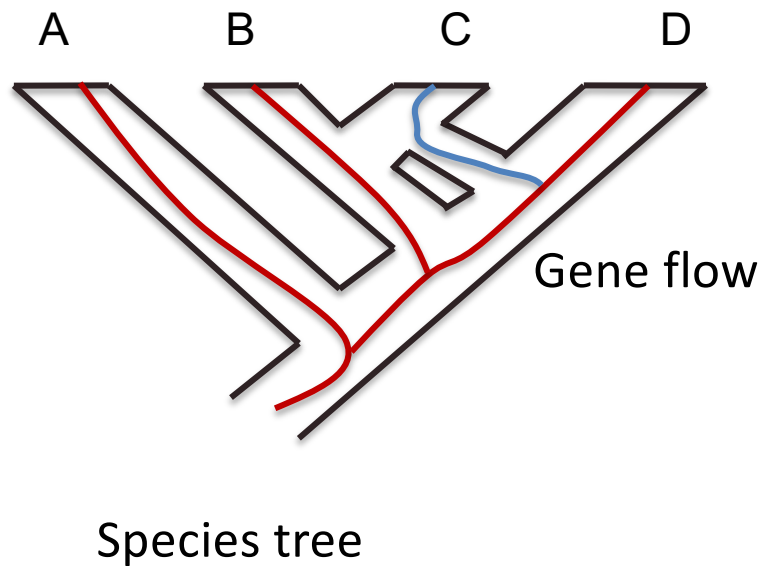
Julia packages for phylogenetic networks

PhyloNetoworks: <https://juliaphylo.github.io/PhyloNetworks.jl/stable/>

SNaQ: <https://juliaphylo.github.io/SNaQ.jl/dev/>

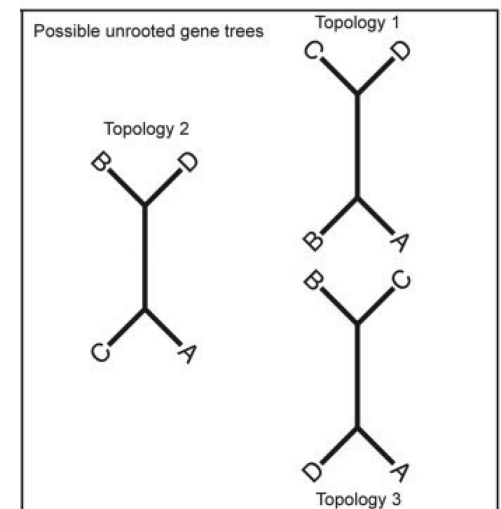
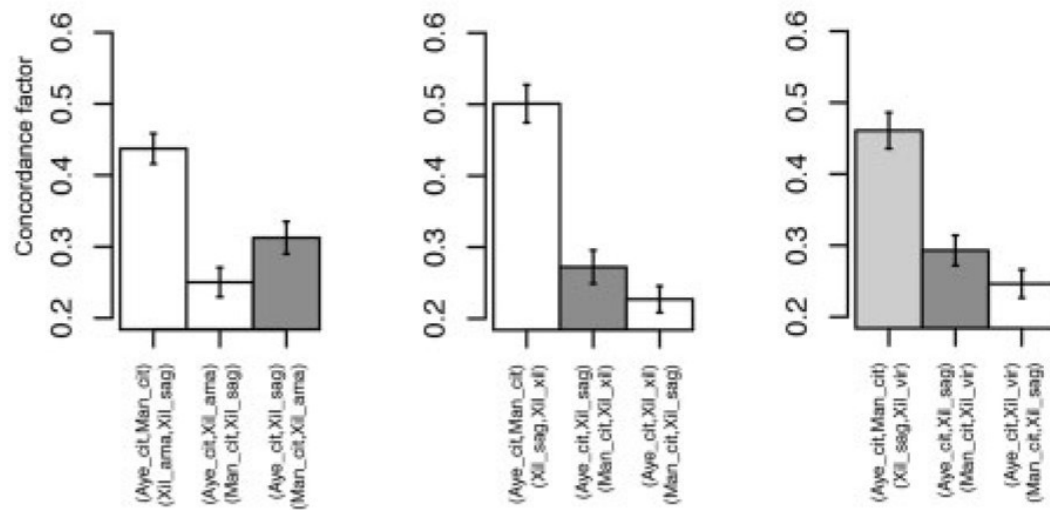
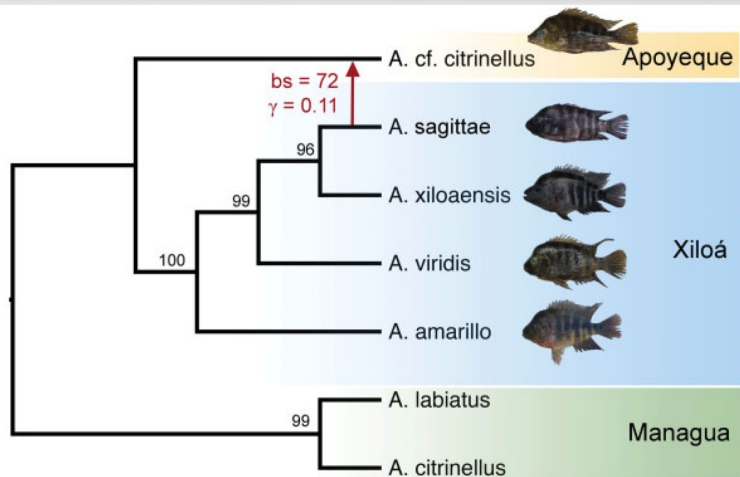
PhyloUtilities: <https://juliaphylo.github.io/PhyloUtilities/>

PhyloPlots: https://juliaphylo.github.io/PhyloPlots.jl/stable/man/getting_started/



PhyloNetworks uses **quartet-based concordance factors** to quantify gene tree discordance

Solis-Lemus et al. (2016) PLoS Genetics



Implementing SNP data in PhyloNetworks



```
← → | 📄 | 💾 | ☐ Source on Save | 🔍 | ✨ | 📝  
1  
2 SNPs2CF(wd=getwd(), seqMatrix,  
3         indels.as.fifth.state=F,  
4         bootstrap=T, boots.rep=100,  
5         outputName="SNPs2CF.csv",  
6         n.quartets="all",  
7         cores=1)  
8
```

Olave y Meyer 2020 Syst Biol
www.github.com/melisaolave/SNPs2CF

Locus →	1	2	3	4	5	6	7	8	9	10
Sp A	1	1	0	1	0	1	1	0	0	1
Sp B	1	1	0	1	0	0	0	1	1	0
Sp C	0	0	1	0	1	1	1	0	1	0
Sp D	0	0	1	0	1	0	0	1	0	1

AB_CD
50%

AC_BD
30%

AD_BC
20%

CF = 0.5

CF = 0.3

CF = 0.2

Phylogenetic networks under multispecies network coalescent (MSNC)

Sequences

Locus 1

```
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
ATCAAAATTTGGCGGC
```

Locus 2

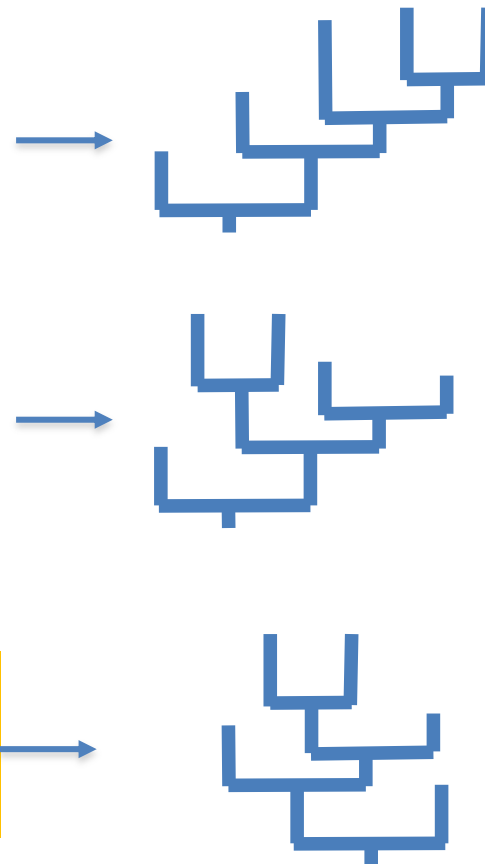
```
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
TTGCAAAGGGGCGC
```

⋮

Locus n

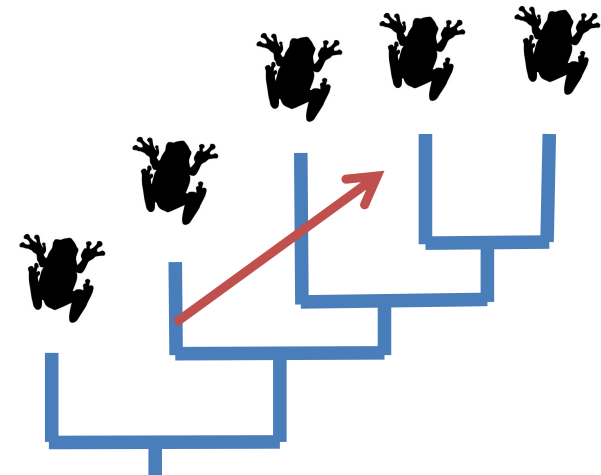
```
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
CCTGCTCCCGGTCGTA
```

Gene trees



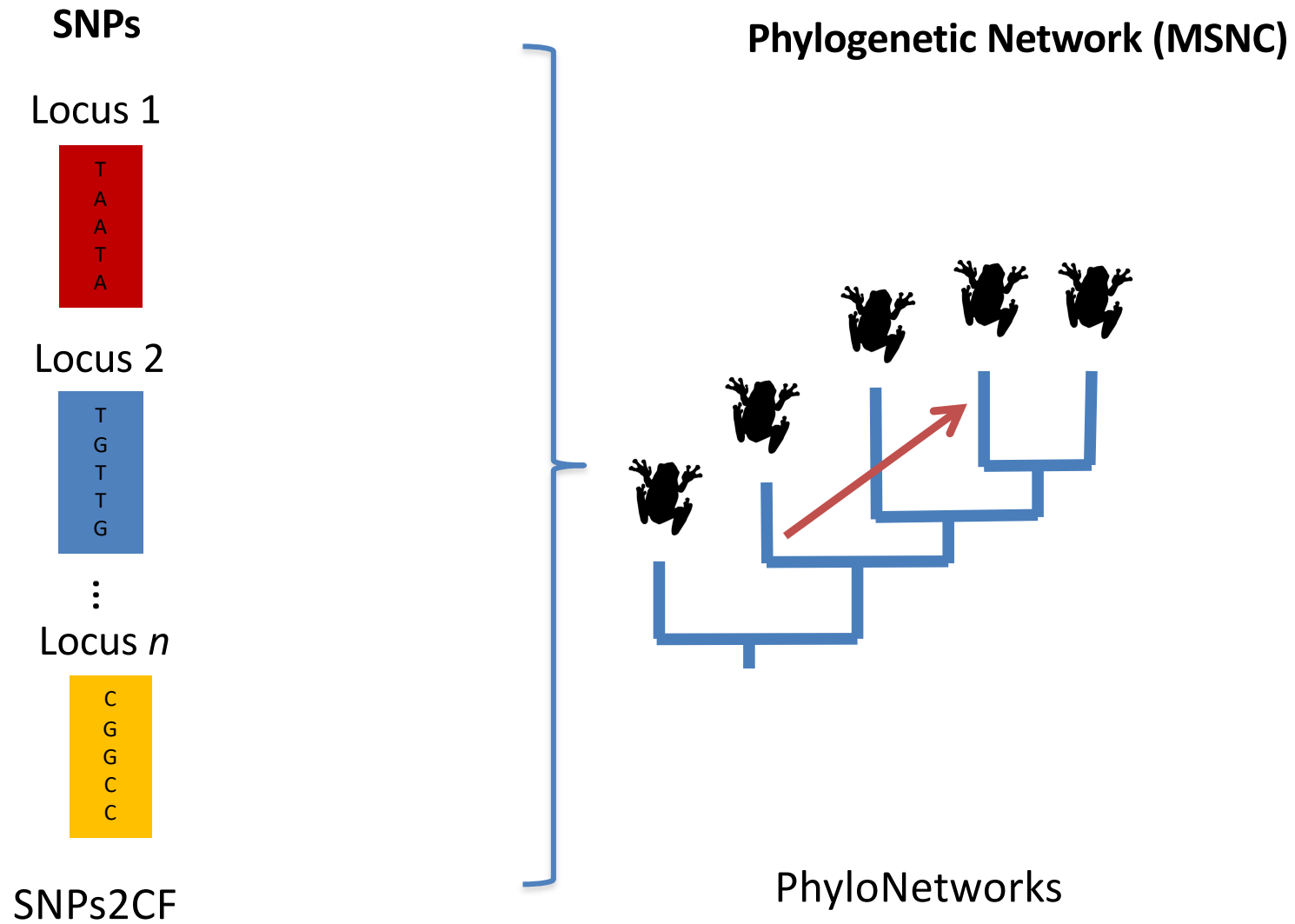
raxml

Phylogenetic Networks (MSNC)



PhyloNetworks

Phylogenetic networks under multispecies network coalescent (MSNC)



Advantages of Using SNPs in Phylogenomics (including Network Reconstruction):

(1) Flexibility

RADseq (short loci) and other technologies that sequence conserved genomic regions (UCEs, hybrid enrichment) may not contain enough mutations for gene tree reconstruction.

Advantages of Using SNPs in Phylogenomics (including Network Reconstruction):

(2) No need for phasing

If SNPs are unlinked, then phasing chromosomes is not necessary, whereas it *is* important for gene tree inference.

This is especially advantageous when working with polyploid species.

Advantages of Using SNPs in Phylogenomics (including Network Reconstruction):

(3) No intralocus recombination issues

Phylogenetic methods (both networks and trees) assume no intralocus recombination, which affects gene tree reconstruction.

If SNPs are sampled at large distances (in base pairs) from one another, this is not a problem.

Advantages of Using SNPs in Phylogenomics (including Network Reconstruction):

(4) Time-saving

Obtaining trees/networks directly from SNPs, without the need to reconstruct gene trees, saves a lot of time!!!

2000 Loci = 2.5 hs (Gene tree) vs. 20-30 seconds (SNPs)

Scenario (t.d.)	Methods	Mean ($\pm$ SD) time taken (in seconds) per gene trees/SNPs			
		500	1000	1500	2000
0.5N	SNPs	2.06	6.01	12.70	21.50
		($\pm$ 0.20)	($\pm$ 0.2)	($\pm$ 0.45)	($\pm$ 0.81)
	GT	2182.23	4436.70	6637.41	8743.46
		($\pm$ 708.70)	($\pm$ 1492.37)	($\pm$ 2226.27)	($\pm$ 2887.18)

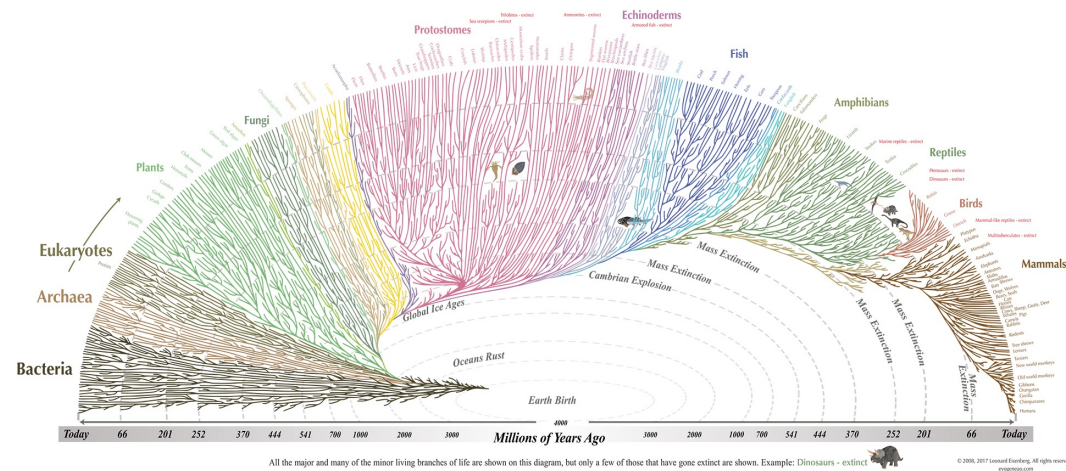


- Network estimation using **gene trees**
 1. Select loci
 2. Separate chromosome phases (recommended)
 3. Reconstruct loci
 4. Infer gene trees
 5. Get Concordance Factors (CF) calculations
 6. Infer Network
- Network estimation using **SNPs**
 1. Select SNPs
 2. Get Concordance Factors (CF) calculations
 3. Infer Network

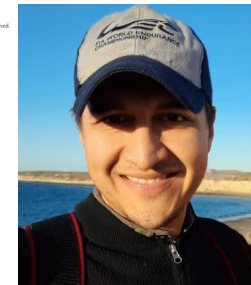
Limitations of Using SNPs in Phylogenomics (including Network Reconstruction):

1. Smaller datasets may mean that SNPs lead to the loss of too much information. ← But solved in new SNPs2CF v 1.7!!!
2. The theoretical framework for working with SNPs is currently much less developed compared to what is known about the behavior of gene trees.
3. The implementation of SNPs2CF in PhyloNetworks does not model differences in mutation types nor does it account for differences in allele frequencies.

Muchas gracias!!!



Our original tutorials (created by Kevin Sanchez):
https://k-sanchez.github.io/workshop_networks_xcbh/



Kevin Sanchez



molave@mendoza-conicet.gob.ar



melisa_olave



melisaolave